

Savar Gupta

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EDUCATION

Simon Fraser University

Bachelor of Applied Science (BASc), Mechatronic Systems Engineering

Burnaby, BC
Sept. 2021 – Sept. 2027 (Expected)

Simon Fraser University, Beedie School of Business

Bachelor of Business Administration (BBA) – Double Degree Program

Burnaby, BC
Sept. 2021 – Sept. 2027 (Expected)

- Relevant coursework: Embedded Systems, Control Systems, PCB Design, Microcontrollers
- Achievements: 4x 1st-place finishes at SFU Beedie ELITE Case Competitions, SFU Beedie Data Analytics Hackathon, and Enactus Canada Western Regionals (2024–2025); RBC Foundation Best Project (Unify) – \$4k (2026); Enactus Canada Nationals 2nd place (2024); Dean's & President's Honour Roll (7x)

EXPERIENCE

TELUS Corporation

Technical Product Manager Co-op, Internet Hardware

Vancouver, BC
Jan. 2026 – Present

- Defined requirements for the **first home Wi-Fi access point to double as a smart-home hub** - an industry first that cuts a standalone \$60 hub from every bundle and turns every internet subscriber into a zero-hardware smart-home upsell
- Tested and validated **Z-Wave** smart-home firmware with ODM partner Arcadyan - **upgrade/downgrade regression cycles**, multi-day **soak/stability testing**, and firmware-version pre-flight checks; reproduced field-reported failures on debug builds, performed **root-cause analysis**, and triaged defects in Jira
- Prototyped on-device **“edge AI”** for home-internet diagnostics - a fine-tuned 1-3B LLM (Ollama) on an AI-accelerator board with self-healing scripts that pinpoint the true fault and auto-resolve it to deflect support calls
- Led a 200-participant, 4-province employee trial for a next-gen **single-box Wi-Fi 7 gateway** - authored install guides, owned device logistics, and cleared firmware/supply-chain blockers to protect the launch date

Unify Social – unifysocial.ca

Co-Founder

Vancouver, BC
July 2024 – Present

- Co-founded Unify, a settlement platform for newcomers to Canada - **350+ users** on a React Native iOS/Android app plus a React web app on a shared Supabase backend
- Built the **RAG-powered AI Assistant**: OpenAI embeddings + **pgvector** retrieval with recency-weighted re-ranking, profile-aware context, SSE token streaming, rate-limiting, and prompt-injection defenses, served via Edge Functions
- Secured **16 partnerships**, ran **25 community events**, and drove **2M+ social views**; designed a **referral revenue model** (3 partners signed) earning commission on partner signups while keeping the app free - roadmap driven by 51 interviews, 225 surveys, and 73 beta testers

Algo Communication Products Ltd.

Electro-Mechanical Engineering Co-op

Burnaby, BC
May 2023 – Aug. 2023

- Diagnosed PCB faults using oscilloscopes and frequency-response analyzers; wrote Python test scripts to **automate fault detection**, increasing **production throughput by 25%** on speaker and horn product lines
- Validated **200+** PCB assemblies and mechanical enclosures for outdoor cameras against design specs, achieving **98% first-pass compliance** across **5 product families**

PROJECTS

Health & Activity Wearable – Hardware & Firmware Engineer | *KiCad, ESP32-S3, C/C++*

May 2026 – Present

- Designed a **2-layer mixed-signal PCB** in KiCad for an ESP32-S3 wearable with IMU, PPG heart-rate sensing, and LiPo power management; completed schematic capture, custom footprints, DRC validation, and fabrication through JLCPCB
- Developed **bare-metal I2C sensor drivers** in Embedded C/C++ for IMU and PPG sensors, scheduled with FreeRTOS tasks and validated communication using a logic analyzer
- Built an **on-device TinyML activity classifier** (Edge Impulse) on the ESP32-S3 and streamed real-time predictions to a companion mobile app over BLE without cloud processing

SFU Rocketry – Propulsion & Mechanical Engineer | *SolidWorks, FEA, Pneumatics*

Oct. 2022 – May 2024

- Developed, validated, and hot-fired LOX/Ethanol liquid rocket engine using SolidWorks and FEA – full design-to-test cycle including thermal and structural analysis
- Designed and manufactured an automated pneumatic valve system for LOX/Ethanol tanks, enabling precise flow regulation

TECHNICAL SKILLS

Product: Roadmapping, PRDs, A/B Testing, User Research & Interviews, Customer Journey Mapping, Financial Modeling, Supply Chain Management, Looker Studio, Tableau, PostHog, Jira, Confluence, Figma

Hardware: C/C++, FreeRTOS, I2C/SPI, Logic Analyzer, KiCad, SolidWorks, MATLAB, ESP32, PCB Design, BLE

Software: TypeScript, JavaScript, Python, React, React Native, Expo, Next.js, Astro, SQL, Git, Worktrees

AI Tooling: Claude Code, Cursor, Codex, OpenRouter, Ollama, RAG pipelines, AI Agents

Cloud & Backend: Supabase (Auth, PostgreSQL, Edge Functions), AWS S3, Cloudflare Workers, Sanity CMS, Resend